

Training a small network

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Updating parameters

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check the mean convention for Training a small network. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Training a small network. Proposed explanation: “Every parameter was trained”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Run a new training case independently

Keep $h = \text{ReLU}(x)$, train only v , and use one pair $(2, 2)$, $v = 0$ and rate 0.125. Use squared error and gradient 2 times $(2v - 2)$ times 2. Perform two updates, reporting each new v and loss. Explain why the second update changes nothing.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the first update

At $v = 0$, calculate both predictions, errors, MSE and gradient, then update at rate 0.1.

My reasoning and evidence

6. W2 Check the mean convention

At $v = 1$, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

My reasoning and evidence

7. W3 Freeze before checking

At final $v = 1.875$, calculate the prediction and squared error on reserved pair (3, 6). May its target be included in the four reported training updates?

My reasoning and evidence

8. W4 Run a new training case independently

Keep $h = \text{ReLU}(x)$, train only v , and use one pair (2, 2), $v = 0$ and rate 0.125. Use squared error and gradient 2 times $(2v - 2)$ times 2. Perform two updates, reporting each new v and loss. Explain why the second update changes nothing.

My reasoning and evidence

9. W5 Expose a frozen representation limit

For the lesson architecture at $x = -1$, give h and prediction for any finite v . If target is -2 , calculate squared error and explain why changing v cannot fix it.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled training a small network evidence sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Training a small network

Train a single output weight while clearly marking the frozen hidden computation.

Toy $h = \text{ReLU}(x)$, $y = vh$. One training pair $x = 3$, target 3; start $v = 0$; squared loss; rate $1/18$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Training a small network

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze hidden step

$h=3$ for this pair; only v is trainable.



2. Compute initial gradient

Loss 9; gradient $2(3v-3) \times 3 = -18$ at $v=0$.



3. Update and verify

v becomes 1; prediction 3 and loss 0. Next gradient is 0.

CHANGE ONE THING

Change target from 3 to 6 and restart $v = 0$ with the same rate.

Explain the changed result using the matching lesson.

Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Train a single output weight while clearly marking the frozen hidden computation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? v becomes 1; prediction 3 and loss 0. Next gradient is 0.

1. Freeze hidden step
2. Update and verify
3. Compute initial gradient

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Update and verify v becomes 1; prediction 3 and loss 0. Next gradient is 0. Only the output weight v is being fitted in this toy network; the hidden ReLU is fixed. The training input gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one training pair is exact fitting of that pair, not evidence about other inputs.

Practice 2: explain the deciding evidence

Changed-case practice: Change target from 3 to 6 and restart $v = 0$ with the same rate. Which conclusion is supported by the supplied facts?

1. First gradient = -36; v becomes 2; prediction 6 and loss 0.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: First gradient=-36; v becomes 2; prediction 6 and loss 0. The hidden value remains 3, but the target residual and required output weight double.

Repair a missing building block

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
